How Do Concept Dependencies Shape Latent Space Geometry?

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Abstract

Understanding how models represent real-world features in their latent space is critical for robust explainability. Many previous works rely on unverified geometric assumptions about Concept Activation Vectors (CAVs), such as vector orthogonality for independent concepts. In this work, we evaluate these hypotheses by calculating CAV cosine similarities using controlled synthetic datasets created using FunnyBirds framework. We confirm that independent concepts are successfully represented by orthogonal directions, achieving a near-zero cosine similarity of 0.03. Conversely, we demonstrate that hierarchical geometric assumptions are not absolute and depend heavily on the experimental setup. Furthermore, systematically reducing the hierarchical correlation strength fails to meaningfully alter the latent angles, with cosine similarities remaining virtually static (e.g., shifting only from 0.87 to 0.82 with color-based concepts, and holding at 0.99 for structural features).

1. Introduction

The importance of ML model explainability has already been evidenced by regulators highlighting it as a necessary step in model development (European Union, 2024, Art. 13). Explaining the outputs of computer vision models often requires researchers to look into the representations of image features in the models' latent space. While the relationships between empirical features in the input datasets are relatively easy to check, the ways in which models understand those relationships and represent them in their latent space are significantly harder to investigate despite being the direct link between the real-world features and model outputs.

While the implications of various assumptions on CAV geometry have been explored in the past (Pahde et al., 2022; Erogullari et al., 2025; Nguyen et al., 2026), the factual existence of such geometric relationships between CAVs hasn't yet been empirically tested. Previous works often rely on unproven assumptions on these relationships. Erogullari et al. (2025) assumes orthogonality of CAVs between independent features, Nguyen et al. (2026) assumes orthogonality between CAVs in the cases where features are hierarchically dependent. Our work fits into this gap by examining the geometrical relations between CAVs in models trained using the Funnybirds (Hesse et al., 2023) dataset through calculating their cosine similarity. The selection of this framework enables the generation of data with strictly controlled attribute relationships, automatically producing bird images defined by interpretable concepts. The experiment pipeline can be seen on Figure 1.

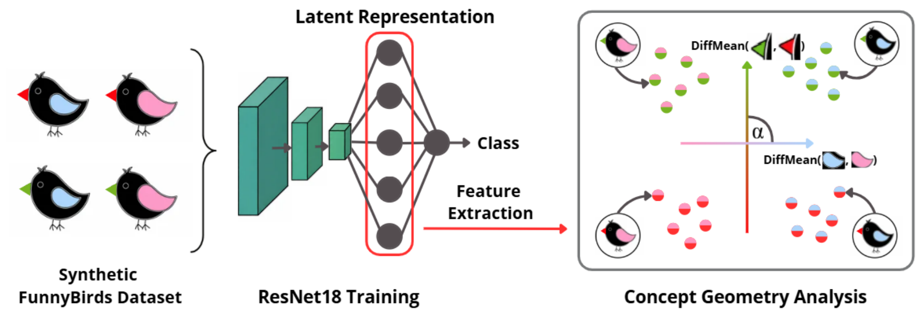

Figure 1: Overview of the experimental pipeline for evaluating latent space geometry under controlled concept dependencies. Synthetic FunnyBirds datasets with predefined concept relationships are used to train a ResNet18 classifier. Mean of difference vectors are extracted from the model's latent representations and compared using cosine similarity.

Our key contributions are:

1. We show that for independent concepts, their corresponding CAVs in the model latent space are orthogonal, empirically verifying the assumption originally made by Erogullari et al. (2025); this finding holds both in a simple space with only two concepts, and in a complex space with over 10 concepts and significant noise in the images.

2. We show that for hierarchical concepts, the geometric assumption (Equation 2) originally made by Nguyen et al. (2026) isn't absolute, with the results depending on the experimental setup.

3. We show that changing the strength of hierarchy between concepts does not proportionally impact the angle between their corresponding CAVs.

2. Methodology

Dataset Generation To support our experiments, we extended the original FunnyBirds framework with our own data generation module, which enables the specification of probabilistic relationships (e.g. independence, hierarchical dependence) between selected concepts and the automatic generation of bird configurations. The generated configurations are then transformed into individual bird images, each of which also includes a randomised camera position and colorful background objects in order to increase image diversity and force the model to learn actual bird features.

Model Training For each generated dataset, we trained a ResNet18 classifier (He et al., 2016) from scratch to predict the generated bird classes from the images. After training, feature vectors from the last hidden layer of the ResNet18 network were extracted and used as latent representations for further analysis.

CAV extraction We use CAVs to represent concepts in the latent space. Following the mean of difference approach, CAVs are computed from latent representations corresponding to different concept values. The resulting vectors are then used to evaluate geometric relationships between concepts.

Evaluation of Geometric Hypoteses The extracted CAVs are used to evaluate geometric assumptions proposed in Erogullari et al. (2025) and Nguyen et al. (2026). Following Erogullari et al. (2025), we test whether independent concepts are represented by orthogonal directions in the latent space:

$$SV_A \perp SV_B, \tag{1}$$

where SV_A denotes the mean of difference CAV of concept A. Following Nguyen et al. (2026), we test whether hierarchical concepts satisfy the proposed geometric relationship

$$(SV_A - SV_B) \perp SV_A, \tag{2}$$

where $A \implies B$. Both hypotheses are evaluated by calculating the cosine similarity between the corresponding concept vectors.

3. Experiments

We evaluated 7 experimental configurations within the FunnyBirds framework, each containing 1000 images per class. We report results for the scenarios testing ideal independence and varying strengths of hierarchical relationships. Full validation accuracies for all trained models are reported in Appendix A, and the corresponding visualizations are presented in Appendix B.

Geometry of Independent Concepts To evaluate the latent behavior of independent concepts, we generated a 4-class dataset, which represented all binary combinations of two features: two beak models and two legs models, with all other features frozen.

Impact of Hierarchical Relationship Strength To investigate how the degree of hierarchical dependency affects latent geometry, we evaluated two distinct dataset generation methods: Method 1, which models color correlations using a 3-class dataset, and Method 2, which models probabilistic concepts with added background noise using a 2-class dataset. In both methods, we systematically reduced the strength of the concept correlation (X) from 100% down to 80% and 60%. We define X as the probability of a specific correlated feature occurring within its designated class.

Class configurations for all three experiments are shown in Table 1. Table 2 explicitly defines the concepts used to extract the Semantic Vectors (denoted as SV_A and SV_B) for each experiment, and reports the resulting cosine similarities across all methods for the varying values of X.

4. Conclusion and Discussion

Our experiments confirm that the extracted vectors for independent concepts are nearly orthogonal, exhibiting a mean cosine similarity of 0.0392. This aligns with an empirical baseline computed from 100,000 random vector pairs in the same space (mean cosine similarity of 0.0001 ± 0.0443), validating the assumption made by Erogullari et al. (2025). When investigating relationships between vectors that describe hierarchically dependent concepts, orthogonality is achieved through Method 2, but not through Method 1. Furthermore, reducing the hierarchical dependence strength did not meaningfully impact the angle between the respective CAVs.

There are four significant limitations to these findings. First, our entire study was performed using a dataset with artificially clear relations between concepts, posing questions about the applicability of these findings to real-life image datasets. Second, the differences in results between different ways of modelling hierarchical relationships between concepts suggest potential issues with our experiment design methods; mediocre classification accuracy suggests that the model may be relying on simple color cues instead of actual structural hierarchies. Third, hierarchical dependencies may inherently form complex, non-linear structures that can't be captured by linear analysis. Fourth, deep ResNets have been found to experience rank collapse just before the classification head (Súkeník et al., 2026), which may reduce the value of findings from the last hidden layer. Future work should validate these geometric findings across diverse architectures and various model layers.

Table 1: Class definitions for all evaluated datasets. Probabilistic features occur with probability $X \in \{100\%, 80\%, 60\%\}$. For Method 1, "else random" implies uniform sampling from the remaining colors.

Experiment	Class	Fixed Features	Probabilistic Feature
Independent	0	beak 0, legs 0	–
	1	beak 0, legs 1	–
	2	beak 1, legs 0	–
	3	beak 1, legs 1	–
Method 1	0	beak 0, legs 0	wings: color 1 (X%), else random
	1	beak 1, legs 0	wings: color 2 (X%), else random
	2	beak 1, legs 1	wings: color 3 (X%), else random
Method 2	0	legs: 0 (50%), 1 (50%)	eyes: 0 (X%), 1 (100-X%)
	1	legs: 2 (50%), 3 (50%)	eyes: 2 (X%), 1 (100-X%)

Table 2: Extracted Semantic Vectors (SV_A, SV_B) computed via DiffMeans, and the resulting latent geometry metrics. Notation like "1 vs. 0" indicates the DiffMeans vector calculated between feature class 1 and 0. For the Independent dataset, correlation strength X does not apply.

Experiment	Extracted Vectors	Metric	100%	80%	60%
Independent	SV_A: beak (1 vs. 0)	Cossim(SV_A, SV_B)	0.0392	–	–
	SV_B: legs (1 vs. 0)	Cossim($SV_A - SV_B$	0.6931	–	–
Method 1	SV_A: beak (1 vs. 0)	Cossim(SV_A, SV_B)	0.8710	0.8357	0.8294
	SV_B: wings color (2 vs. 1)	Cossim($SV_A - SV_B$, SV_A)	0.2539	0.2867	0.2921
Method 2	SV_A: eyes (2 vs. 0)	Cossim(SV_A, SV_B)	0.9999	1.0000	0.9999
	SV_B: legs (2 vs. 0)	Cossim($SV_A - SV_B$, SV_A)	0.0050	0.0037	0.0057

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Appendix A. Validation Accuracies

Table 3 shows the validation accuracies for all trained models.

Table 3: Validation accuracies for all trained models.

Experiment	Correlation Strength (X)	Validation Accuracy (%)
Independent	–	93.13%
Method 1	100%	99.00%
Method 1	80%	88.83%
Method 1	60%	88.33%
Method 2	100%	95.25%
Method 2	80%	94.25%
Method 2	60%	95.25%

Appendix B. Visualizations of Latent Space Geometry

This appendix provides PCA projections of the latent representations for all evaluated experimental configurations. Each figure visualizes the distribution of image representations (colored points) alongside the extracted Semantic Vectors (arrows) corresponding to the evaluated concepts.

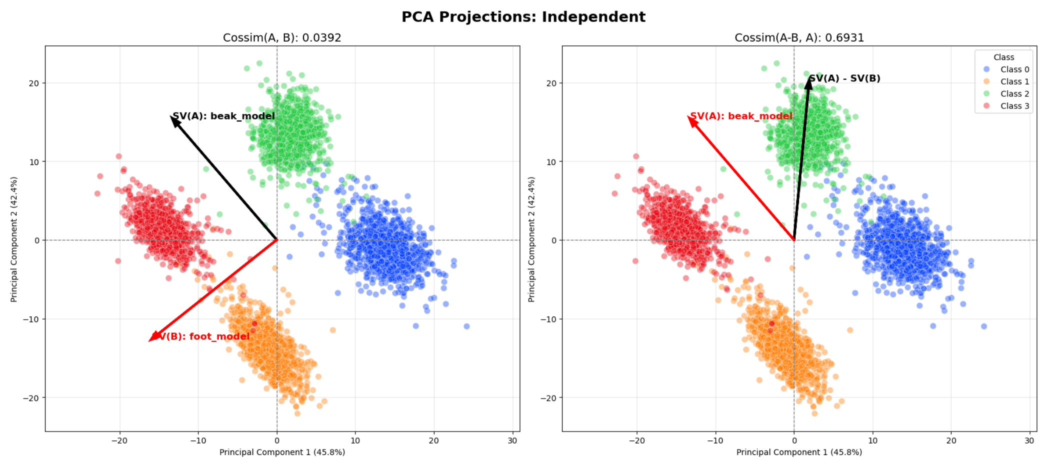

Figure 2: PCA projections of latent representations for the Independent concepts dataset. Points represent individual images colored by class, with arrows indicating the extracted Semantic Vectors SV_A (beak model) and SV_B (foot model). The near-orthogonal angle between the vectors (Cossim = 0.0392) visually confirms the assumption that independent concepts form orthogonal directions in the latent space.

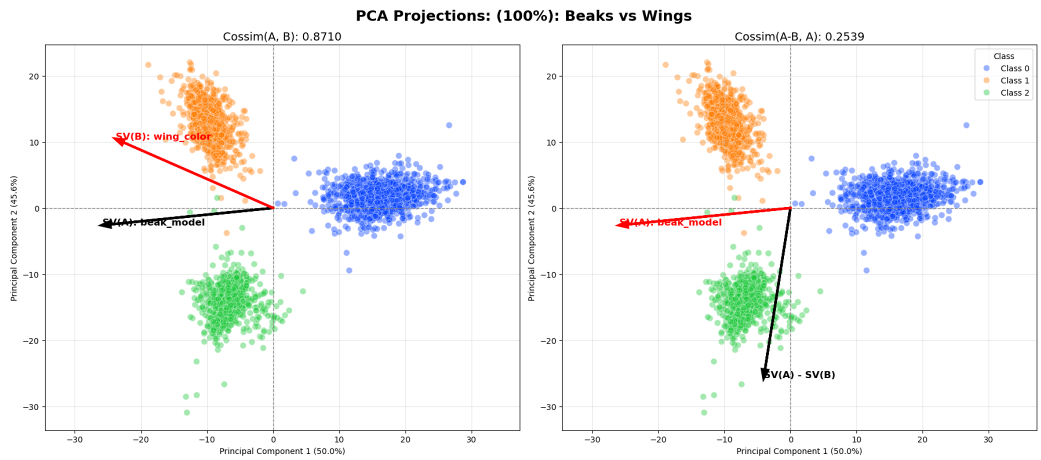

Figure 3: PCA projections for Method 1 (beak vs. wings color) at correlation strength $X = 100\%$. The left plot shows the relation between SV_A and SV_B, while the right plot evaluates the HCEP assumption ($SV_A - SV_B$). The high cosine similarity (0.8710) indicates that strictly correlated color concepts are highly entangled in the latent space, failing to achieve the expected orthogonality.

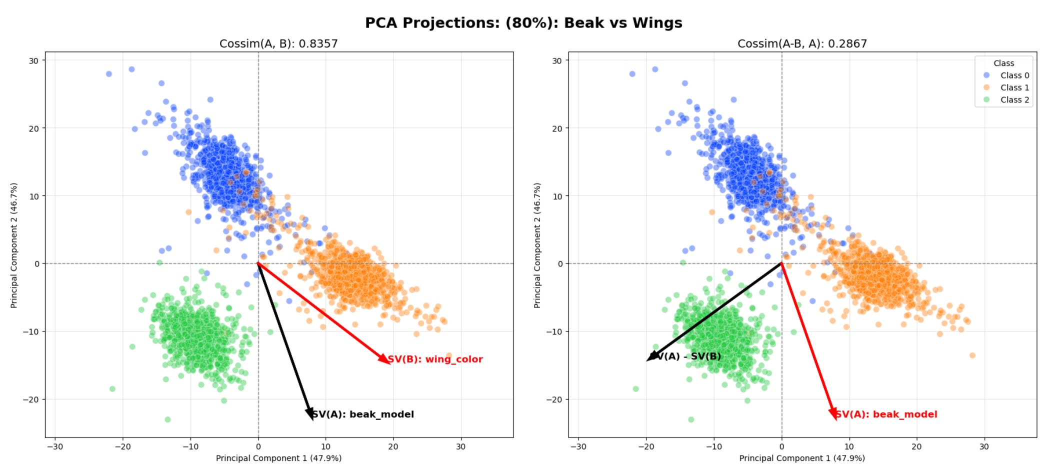

Figure 4: PCA projections for Method 1 (beak vs. wings color) at correlation strength $X = 80\%$. Lowering the correlation strength introduces more variance and slightly separates the point clusters, but the semantic vectors remain highly aligned (Cossim = 0.8357), showing that the model still heavily associates the two concepts.

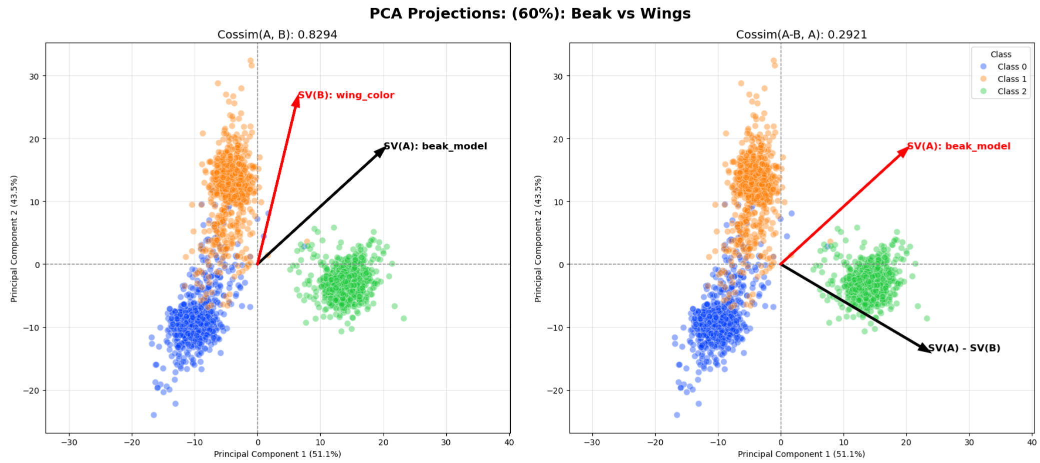

Figure 5: PCA projections for Method 1 (beak vs. wings color) at correlation strength $X = 60\%$. Further reducing the correlation strength does not meaningfully separate the concept vectors (Cossim = 0.8294), demonstrating that decreasing hierarchical dependence strength does not proportionally impact the angle between the respective CAVs.

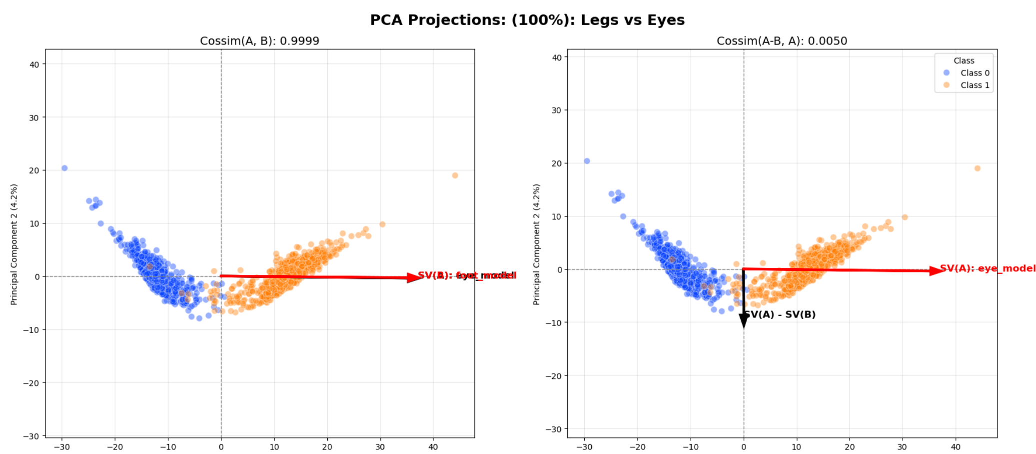

Figure 6: PCA projections for Method 2 (legs vs. eyes with background noise) at correlation strength $X = 100\%$. The left plot shows SV_A and SV_B are nearly collinear, but the right plot shows the difference vector $(SV_A - SV_B)$ is strictly orthogonal to SV_A (Cossim = 0.0050). This visually confirms that the HCEP geometric assumption holds true for this specific structural setup.

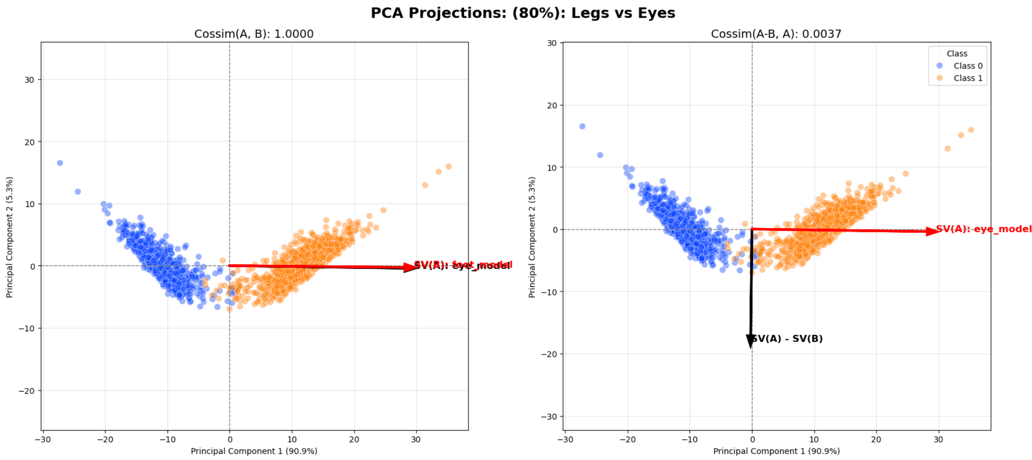

Figure 7: PCA projections for Method 2 (legs vs. eyes with background noise) at correlation strength $X = 80\%$. The geometric structure and strict orthogonality of the difference vector (Cossim = 0.0037) are preserved despite the added probabilistic noise in the features.

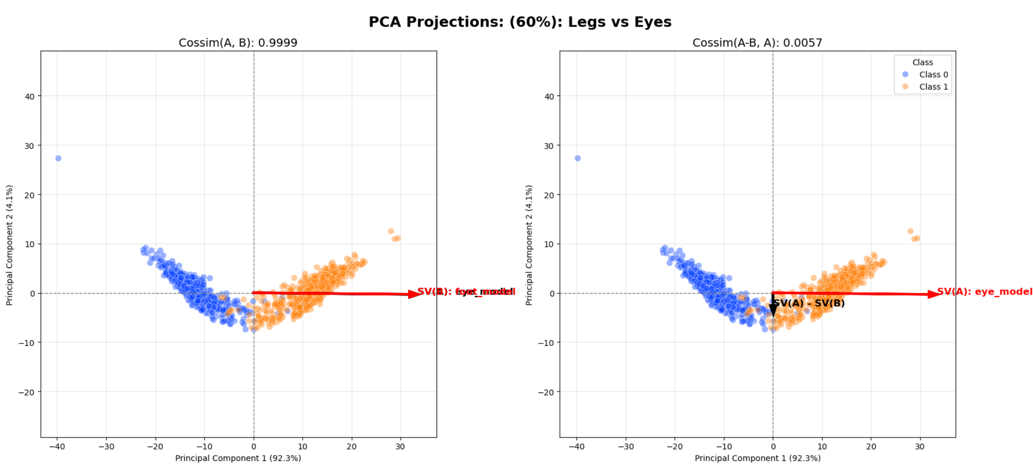

Figure 8: PCA projections for Method 2 (legs vs. eyes with background noise) at correlation strength $X = 60\%$. The latent geometry remains highly stable (Cossim of the difference vector = 0.0057), demonstrating that for these structural features, correlation strength does not impact the strict geometric relationship between the vectors.